

Miscanthus: EDA and Community Diversity Report

Bolívar Aponte Rolón

2025-11-18

About this report

This Quarto report executes two existing analysis scripts:

- `04_eda.R` — Exploratory Data Analysis of Microbial Communities
- `05_alpha_beta_div.R` — Community Diversity Analyses

Both scripts assume project-root working directory (for sourcing `R/utls/00_setup.R`). Place this `.qmd` at the repository root so relative paths resolve as expected.

If your environment needs it, ensure all packages used by the scripts are installed before rendering.

Note

Render from the repository root:

- CLI: `quarto render mxg_eda_report.qmd --execute-dir .`
- R: `quarto::quarto_render("mxg_eda_report.qm")`

Section 1 — Exploratory Data Analysis

This section runs the full workflow in [04_eda.R](#), including:

- Dataset exploration (metadata, taxonomic distribution)
- Read count summaries and visualizations
- Good's coverage estimates
- ASV prevalence/core taxa summaries

```
source(
  "R/04_eda.R",
  local = knitr::knit_global(),
  echo = TRUE,
  print.eval = TRUE
)
```

```
> source("R/utills/00_setup.R")
```

```
> str(main_mxg_physeq_list, max.level = 2)
```

```
List of 3
```

```
$ mxg_ef          :List of 2
```

```
..$ ef_16S_physeq:Formal class 'phyloseq' [package "phyloseq"] with 5 slots
```

```
..$ ef_AMF_physeq:Formal class 'phyloseq' [package "phyloseq"] with 5 slots
```

```
$ mxg_lamps_2018:List of 2
```

```
..$ lamps_2018_16S_physeq:Formal class 'phyloseq' [package "phyloseq"] with 5 slots
```

```
..$ lamps_2018_ITS_physeq:Formal class 'phyloseq' [package "phyloseq"] with 5 slots
```

```
$ mxg_lamps_2022:List of 2
```

```
..$ lamps_2022_16S_physeq:Formal class 'phyloseq' [package "phyloseq"] with 5 slots
```

```
..$ lamps_2022_AMF_physeq:Formal class 'phyloseq' [package "phyloseq"] with 5 slots
```

```
> names(main_mxg_physeq_list)
```

```
[1] "mxg_ef"          "mxg_lamps_2018" "mxg_lamps_2022"
```

```
> length(main_mxg_physeq_list)
```

```
[1] 3
```

```
> purrr::iwalk(main_mxg_physeq_list, function(project_list,
```

```
+   project_name) {
```

```
+   cat("### PROJECT:", project_name, "###\n")
```

```
+   purrr::iwalk .... [TRUNCATED]
```

```
### PROJECT: mxg_ef ###
```

```
=== Analysis for: mxg_ef_ef_16S_physeq ===
```

```
OTU table dimensions: 15182 223
```

```
Number of taxa: 15182
```

```
Number of samples: 223
```

```
Raw count summary:
```

```
Min. 1st Qu. Median Mean 3rd Qu. Max.
```

0.000 0.000 0.000 1.052 0.000 4609.000

=== Analysis for: mxg_ef_ef_AMF_physeq ===

OTU table dimensions: 1208 200

Number of taxa: 1208

Number of samples: 200

Raw count summary:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00	0.00	0.00	19.27	0.00	46148.00

PROJECT: mxg_lamps_2018

=== Analysis for: mxg_lamps_2018_lamps_2018_16S_physeq ===

OTU table dimensions: 49404 788

Number of taxa: 49404

Number of samples: 788

Raw count summary:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000	0.000	0.000	0.323	0.000	8553.000

=== Analysis for: mxg_lamps_2018_lamps_2018_ITS_physeq ===

OTU table dimensions: 6941 93

Number of taxa: 6941

Number of samples: 93

Raw count summary:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000	0.000	0.000	4.691	0.000	13189.000

PROJECT: mxg_lamps_2022

=== Analysis for: mxg_lamps_2022_lamps_2022_16S_physeq ===

OTU table dimensions: 3939 12

Number of taxa: 3939

Number of samples: 12

Raw count summary:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00	0.00	0.00	12.26	0.00	4952.00

=== Analysis for: mxg_lamps_2022_lamps_2022_AMF_physeq ===

OTU table dimensions: 200 11

Number of taxa: 200

Number of samples: 11

Raw count summary:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0	0.0	0.0	100.9	0.0	15092.0

```
> physeq_summary <- purrr::imap_dfr(main_mxg_physeq_list,
+   function(project_list, project_name) {
+     purrr::imap_dfr(project_list, functi .... [TRUNCATED]

> physeq_summary
  project_list      project_name region n_taxa n_samples total_reads
1      mxg_ef      ef_16S_physeq   16S  15182      223    3560545
2      mxg_ef      ef_AMF_physeq   AMF   1208      200    4656452
3 mxg_lamps_2018 lamps_2018_16S_physeq 16S 49404      788   12590589
4 mxg_lamps_2018 lamps_2018_ITS_physeq ITS  6941       93    3028081
5 mxg_lamps_2022 lamps_2022_16S_physeq 16S  3939       12     579259
6 mxg_lamps_2022 lamps_2022_AMF_physeq AMF   200       11     222004
  min_reads_per_sample max_reads_per_sample mean_reads_per_sample
1                0          127114          15966.57
2                1          175072          23282.26
3                0           55112          15977.90
4           4597          103732          32560.01
5          20339           73953          48271.58
6           1868           78623          20182.18
  median_reads_per_sample
1          11504.0
2          17227.0
3          13381.5
4          24912.0
5          46876.0
6          13102.0
```

```
> nested_summary <- explore_nested_phyloseq(main_mxg_physeq_list)
```

```
=====
PROJECT: MXG_EF
=====
```

```
-----
Sequencing Type: EF_16S_PHYSEQ
-----
```

```
Basic Summary:
```

	Parameter	Phys1
1	Average number of reads per OTU	234.5241
2	Average number of reads per sample	15966.5695
3	Number of OTUs	15182.0000
4	Number of samples	223.0000

5 Total number of reads 3560545.0000

Read/Sequencing Summary:

[[1]]

[1] "1] Min. number of reads = 0"

[[2]]

[1] "2] Max. number of reads = 127114"

[[3]]

[1] "3] Total number of reads = 3560545"

[[4]]

[1] "4] Average number of reads = 15966.5695067265"

[[5]]

[1] "5] Median number of reads = 11504"

[[6]]

[1] "7] Sparsity = 0.971745511707574"

[[7]]

[1] "6] Any OTU sum to 1 or less? YES"

[[8]]

[1] "8] Number of singletons = 28"

[[9]]

[1] "9] Percent of OTUs that are singletons \n (i.e. exactly one read detected across

[[10]]

[1] "10] Number of sample variables are: 15"

[[11]]

[1] "crop"	"plot_description"	"plot_id"
[4] "subblock"	"treatment"	"sample_date"
[7] "timing"	"rep"	"mr_location"
[10] "mr_loc_id"	"sample_description"	"soil_type"
[13] "dk_plotid"	"soil_type_1"	"fertilizer"

Sequencing Type: EF_AMF_PHYSEQ

Basic Summary:

	Parameter	Phys1
1	Average number of reads per OTU	3854.679
2	Average number of reads per sample	23282.260
3	Number of OTUs	1208.000
4	Number of samples	200.000
5	Total number of reads	4656452.000

Read/Sequencing Summary:

[[1]]

[1] "1] Min. number of reads = 1"

[[2]]

[1] "2] Max. number of reads = 175072"

[[3]]

[1] "3] Total number of reads = 4656452"

[[4]]

[1] "4] Average number of reads = 23282.26"

[[5]]

[1] "5] Median number of reads = 17227"

[[6]]

[1] "7] Sparsity = 0.972077814569536"

[[7]]

[1] "6] Any OTU sum to 1 or less? YES"

[[8]]

[1] "8] Number of singletons = 2"

[[9]]

[1] "9] Percent of OTUs that are singletons \n (i.e. exactly one read detected across

[[10]]

[1] "10] Number of sample variables are: 15"

```
[[11]]
[1] "crop"          "plot_description" "plot_id"
[4] "subblock"      "treatment"       "sample_date"
[7] "timing"         "rep"             "mr_location"
[10] "mr_loc_id"     "sample_description" "soil_type"
[13] "dk_plotid"     "soil_type_1"     "fertilizer"
```

```
= = = = =
PROJECT: MXG_LAMPS_2018
= = = = =
```

```
- - - - -
Sequencing Type: LAMPS_2018_16S_PHYSEQ
- - - - -
```

Basic Summary:

	Parameter	Phys1
1	Average number of reads per OTU	2.548496e+02
2	Average number of reads per sample	1.597790e+04
3	Number of OTUs	4.940400e+04
4	Number of samples	7.880000e+02
5	Total number of reads	1.259059e+07

Read/Sequencing Summary:

```
[[1]]
[1] "1] Min. number of reads = 0"

[[2]]
[1] "2] Max. number of reads = 55112"

[[3]]
[1] "3] Total number of reads = 12590589"

[[4]]
[1] "4] Average number of reads = 15977.904822335"

[[5]]
[1] "5] Median number of reads = 13381.5"

[[6]]
```

```

[1] "7] Sparsity = 0.990797642929096"

[[7]]
[1] "6] Any OTU sum to 1 or less? YES"

[[8]]
[1] "8] Number of singletons = 33"

[[9]]
[1] "9] Percent of OTUs that are singletons \n          (i.e. exactly one read detected across

[[10]]
[1] "10] Number of sample variables are: 14"

[[11]]
[1] "sequence_id"          "sample_id"
[3] "crop"                  "year"
[5] "plot"                  "nitrogen_conc"
[7] "replicate"             "sample_date"
[9] "days_after_fertilization" "nucleotide"
[11] "target_region"         "fertilization"
[13] "soil_type"             "project"

-----
Sequencing Type: LAMPS_2018_ITS_PHYSEQ
-----

Basic Summary:

      Parameter      Phys1
1  Average number of reads per OTU    436.26
2 Average number of reads per sample  32560.01
3           Number of OTUs    6941.00
4           Number of samples    93.00
5           Total number of reads 3028081.00

Read/Sequencing Summary:

[[1]]
[1] "1] Min. number of reads = 4597"

[[2]]
[1] "2] Max. number of reads = 103732"

```



```

[[3]]
[1] "3] Total number of reads = 3028081"

[[4]]
[1] "4] Average number of reads = 32560.0107526882"

[[5]]
[1] "5] Median number of reads = 24912"

[[6]]
[1] "7] Sparsity = 0.838925010030782"

[[7]]
[1] "6] Any OTU sum to 1 or less? YES"

[[8]]
[1] "8] Number of singletons = 72"

[[9]]
[1] "9] Percent of OTUs that are singletons \n          (i.e. exactly one read detected across

[[10]]
[1] "10] Number of sample variables are: 10"

[[11]]
[1] "sequence_id"          "sample_id"
[3] "crop"                 "year"
[5] "plot"                 "nitrogen_conc"
[7] "days_after_fertilization" "nucleotide"
[9] "target_region"        "project"

```

```

= = = = =
PROJECT: MXG_LAMPS_2022
= = = = =

```

```

- - - - -
Sequencing Type: LAMPS_2022_16S_PHYSEQ
- - - - -

```

Basic Summary:

	Parameter	Phys1
1	Average number of reads per OTU	147.0574

2	Average number of reads per sample	48271.5833
3	Number of OTUs	3939.0000
4	Number of samples	12.0000
5	Total number of reads	579259.0000

Read/Sequencing Summary:

[[1]]

[1] "1] Min. number of reads = 20339"

[[2]]

[1] "2] Max. number of reads = 73953"

[[3]]

[1] "3] Total number of reads = 579259"

[[4]]

[1] "4] Average number of reads = 48271.5833333333"

[[5]]

[1] "5] Median number of reads = 46876"

[[6]]

[1] "7] Sparsity = 0.782897520521283"

[[7]]

[1] "6] Any OTU sum to 1 or less? YES"

[[8]]

[1] "8] Number of singletons = 5"

[[9]]

[1] "9] Percent of OTUs that are singletons \n (i.e. exactly one read detected across

[[10]]

[1] "10] Number of sample variables are: 6"

[[11]]

[1] "sequence_id" "sample_id" "site" "crop" "replicate"

[6] "year"

Sequencing Type: LAMPS_2022_AMF_PHYSEQ

Basic Summary:

	Parameter	Phys1
1	Average number of reads per OTU	1110.02
2	Average number of reads per sample	20182.18
3	Number of OTUs	200.00
4	Number of samples	11.00
5	Total number of reads	222004.00

Read/Sequencing Summary:

[[1]]

[1] "1] Min. number of reads = 1868"

[[2]]

[1] "2] Max. number of reads = 78623"

[[3]]

[1] "3] Total number of reads = 222004"

[[4]]

[1] "4] Average number of reads = 20182.1818181818"

[[5]]

[1] "5] Median number of reads = 13102"

[[6]]

[1] "7] Sparsity = 0.853636363636364"

[[7]]

[1] "6] Any OTU sum to 1 or less? NO"

[[8]]

[1] "8] Number of singletons = 0"

[[9]]

[1] "9] Percent of OTUs that are singletons \n (i.e. exactly one read detected across

[[10]]

[1] "10] Number of sample variables are: 6"

```

[[11]]
[1] "sequence_id" "sample_id"   "site"         "crop"         "replicate"
[6] "year"

> nested_summary
$mxg_ef
$mxg_ef$ef_16S_physeq
$mxg_ef$ef_16S_physeq$project
[1] "mxg_ef"

$mxg_ef$ef_16S_physeq$region
[1] "16S"

$mxg_ef$ef_16S_physeq$n_samples
[1] 223

$mxg_ef$ef_16S_physeq$n_taxa
[1] 15182

$mxg_ef$ef_16S_physeq$sample_vars
  [1] "crop"           "plot_description" "plot_id"
  [4] "subblock"       "treatment"       "sample_date"
  [7] "timing"         "rep"             "mr_location"
 [10] "mr_loc_id"      "sample_description" "soil_type"
 [13] "dk_plotid"      "soil_type_1"     "fertilizer"

$mxg_ef$ef_16S_physeq$rank_names
[1] "sequence" "kingdom"  "phylum"  "class"    "order"    "family"   "genus"

$mxg_ef$ef_16S_physeq$total_reads
[1] 3560545

$mxg_ef$ef_16S_physeq$min_reads_per_sample
[1] 0

$mxg_ef$ef_16S_physeq$max_reads_per_sample
[1] 127114

$mxg_ef$ef_16S_physeq$mean_reads_per_sample
[1] 15966.57

```

```
$mxg_ef$ef_16S_physeq$median_reads_per_sample  
[1] 11504
```

```
$mxg_ef$ef_AMF_physeq  
$mxg_ef$ef_AMF_physeq$project  
[1] "mxg_ef"
```

```
$mxg_ef$ef_AMF_physeq$region  
[1] "AMF"
```

```
$mxg_ef$ef_AMF_physeq$n_samples  
[1] 200
```

```
$mxg_ef$ef_AMF_physeq$n_taxa  
[1] 1208
```

```
$mxg_ef$ef_AMF_physeq$sample_vars  
[1] "crop" "plot_description" "plot_id"  
[4] "subblock" "treatment" "sample_date"  
[7] "timing" "rep" "mr_location"  
[10] "mr_loc_id" "sample_description" "soil_type"  
[13] "dk_plotid" "soil_type_1" "fertilizer"
```

```
$mxg_ef$ef_AMF_physeq$rank_names  
[1] "sequence" "kingdom" "phylum" "class" "order" "family" "genus"  
[8] "species"
```

```
$mxg_ef$ef_AMF_physeq$total_reads  
[1] 4656452
```

```
$mxg_ef$ef_AMF_physeq$min_reads_per_sample  
[1] 1
```

```
$mxg_ef$ef_AMF_physeq$max_reads_per_sample  
[1] 175072
```

```
$mxg_ef$ef_AMF_physeq$mean_reads_per_sample  
[1] 23282.26
```

```
$mxg_ef$ef_AMF_physeq$median_reads_per_sample  
[1] 17227
```

```

$mxg_lamps_2018
$mxg_lamps_2018$lamps_2018_16S_physeq
$mxg_lamps_2018$lamps_2018_16S_physeq$project
[1] "mxg_lamps_2018"

$mxg_lamps_2018$lamps_2018_16S_physeq$region
[1] "16S"

$mxg_lamps_2018$lamps_2018_16S_physeq$n_samples
[1] 788

$mxg_lamps_2018$lamps_2018_16S_physeq$n_taxa
[1] 49404

$mxg_lamps_2018$lamps_2018_16S_physeq$sample_vars
[1] "sequence_id"          "sample_id"
[3] "crop"                 "year"
[5] "plot"                  "nitrogen_conc"
[7] "replicate"             "sample_date"
[9] "days_after_fertilization" "nucleotide"
[11] "target_region"         "fertilization"
[13] "soil_type"             "project"

$mxg_lamps_2018$lamps_2018_16S_physeq$rank_names
[1] "sequence" "kingdom" "phylum" "class" "order" "family" "genus"

$mxg_lamps_2018$lamps_2018_16S_physeq$total_reads
[1] 12590589

$mxg_lamps_2018$lamps_2018_16S_physeq$min_reads_per_sample
[1] 0

$mxg_lamps_2018$lamps_2018_16S_physeq$max_reads_per_sample
[1] 55112

$mxg_lamps_2018$lamps_2018_16S_physeq$mean_reads_per_sample
[1] 15977.9

$mxg_lamps_2018$lamps_2018_16S_physeq$median_reads_per_sample
[1] 13381.5

```

```

$mxg_lamps_2018$lamps_2018_ITS_physeq
$mxg_lamps_2018$lamps_2018_ITS_physeq$project
[1] "mxg_lamps_2018"

$mxg_lamps_2018$lamps_2018_ITS_physeq$region
[1] "ITS"

$mxg_lamps_2018$lamps_2018_ITS_physeq$n_samples
[1] 93

$mxg_lamps_2018$lamps_2018_ITS_physeq$n_taxa
[1] 6941

$mxg_lamps_2018$lamps_2018_ITS_physeq$sample_vars
  [1] "sequence_id"          "sample_id"
  [3] "crop"                  "year"
  [5] "plot"                  "nitrogen_conc"
  [7] "days_after_fertilization" "nucleotide"
  [9] "target_region"         "project"

$mxg_lamps_2018$lamps_2018_ITS_physeq$rank_names
  [1] "sequence"  "domain"    "phylum"  "class"    "order"
  [6] "family"    "genus"     "species"   "amf.guild" "trophic"
 [11] "guild"     "trait"     "confidence" "notes"    "citation"

$mxg_lamps_2018$lamps_2018_ITS_physeq$total_reads
[1] 3028081

$mxg_lamps_2018$lamps_2018_ITS_physeq$min_reads_per_sample
[1] 4597

$mxg_lamps_2018$lamps_2018_ITS_physeq$max_reads_per_sample
[1] 103732

$mxg_lamps_2018$lamps_2018_ITS_physeq$mean_reads_per_sample
[1] 32560.01

$mxg_lamps_2018$lamps_2018_ITS_physeq$median_reads_per_sample
[1] 24912

```

```

$mxg_lamps_2022
$mxg_lamps_2022$lamps_2022_16S_physeq
$mxg_lamps_2022$lamps_2022_16S_physeq$project
[1] "mxg_lamps_2022"

$mxg_lamps_2022$lamps_2022_16S_physeq$region
[1] "16S"

$mxg_lamps_2022$lamps_2022_16S_physeq$n_samples
[1] 12

$mxg_lamps_2022$lamps_2022_16S_physeq$n_taxa
[1] 3939

$mxg_lamps_2022$lamps_2022_16S_physeq$sample_vars
[1] "sequence_id" "sample_id" "site" "crop" "replicate"
[6] "year"

$mxg_lamps_2022$lamps_2022_16S_physeq$rank_names
[1] "Kingdom" "Phylum" "Class" "Order" "Family" "Genus"

$mxg_lamps_2022$lamps_2022_16S_physeq$total_reads
[1] 579259

$mxg_lamps_2022$lamps_2022_16S_physeq$min_reads_per_sample
[1] 20339

$mxg_lamps_2022$lamps_2022_16S_physeq$max_reads_per_sample
[1] 73953

$mxg_lamps_2022$lamps_2022_16S_physeq$mean_reads_per_sample
[1] 48271.58

$mxg_lamps_2022$lamps_2022_16S_physeq$median_reads_per_sample
[1] 46876

$mxg_lamps_2022$lamps_2022_AMF_physeq
$mxg_lamps_2022$lamps_2022_AMF_physeq$project
[1] "mxg_lamps_2022"

$mxg_lamps_2022$lamps_2022_AMF_physeq$region
[1] "AMF"

```



```

$mxg_lamps_2022$lamps_2022_AMF_physeq$n_samples
[1] 11

$mxg_lamps_2022$lamps_2022_AMF_physeq$n_taxa
[1] 200

$mxg_lamps_2022$lamps_2022_AMF_physeq$sample_vars
[1] "sequence_id" "sample_id"   "site"           "crop"           "replicate"
[6] "year"

$mxg_lamps_2022$lamps_2022_AMF_physeq$rank_names
[1] "Kingdom" "Phylum" "Class"     "Order"     "Family"    "Genus"     "Species"

$mxg_lamps_2022$lamps_2022_AMF_physeq$total_reads
[1] 222004

$mxg_lamps_2022$lamps_2022_AMF_physeq$min_reads_per_sample
[1] 1868

$mxg_lamps_2022$lamps_2022_AMF_physeq$max_reads_per_sample
[1] 78623

$mxg_lamps_2022$lamps_2022_AMF_physeq$mean_reads_per_sample
[1] 20182.18

$mxg_lamps_2022$lamps_2022_AMF_physeq$median_reads_per_sample
[1] 13102

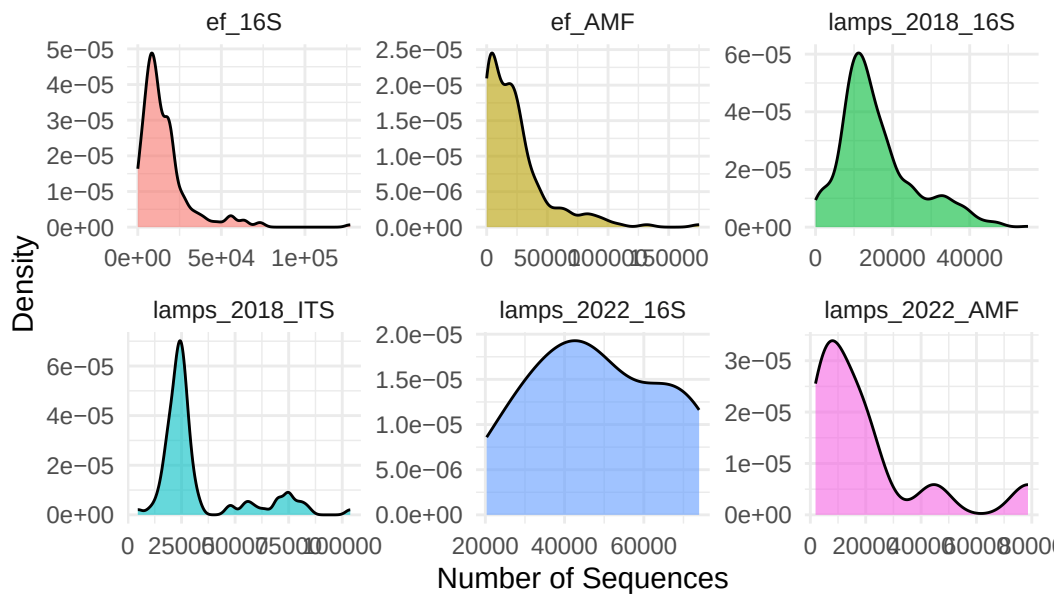

> nested_read_counts <- analyze_read_counts(main_mxg_physeq_list)

> read_count_plots <- list(density = ggplot(nested_read_counts,
+   aes(x = n_seqs, fill = project)) + geom_density(alpha = 0.6,
+   position = .... [TRUNCATED]

> print(read_count_plots)
$density

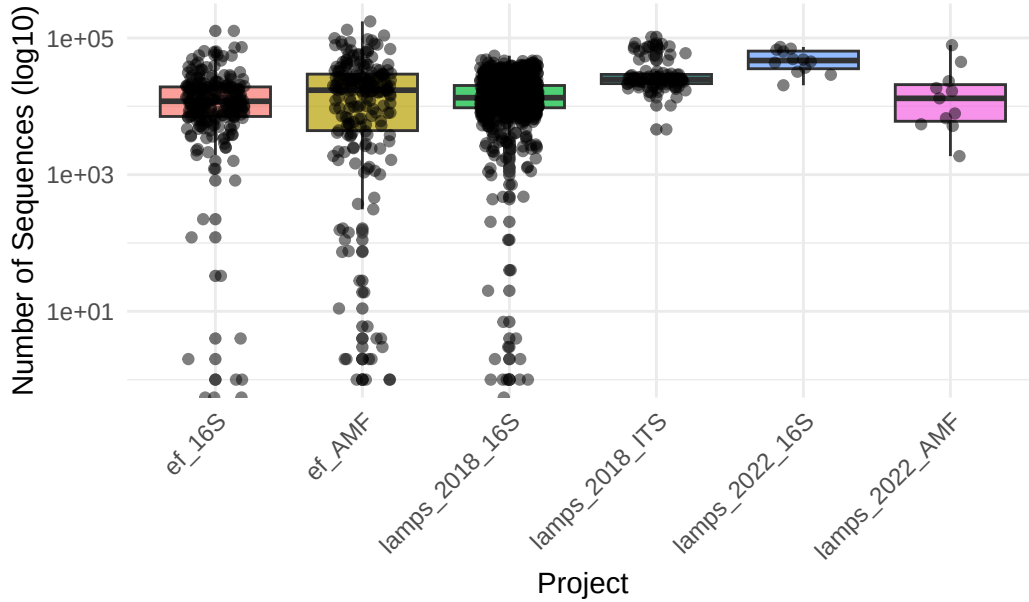
```

Read Count Distribution by Project

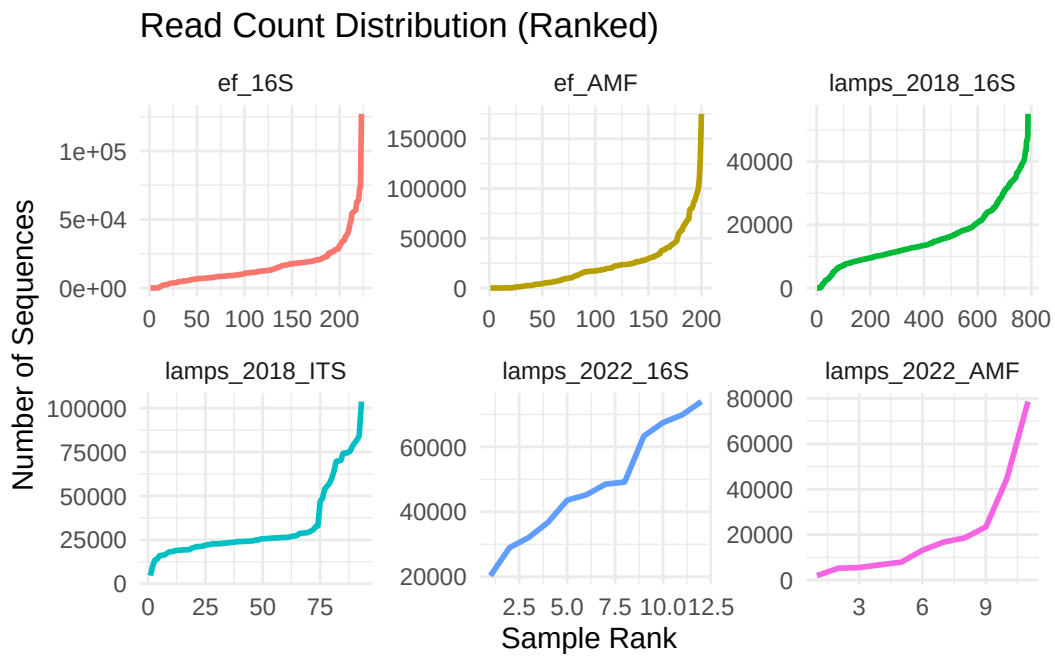


`$boxplot`

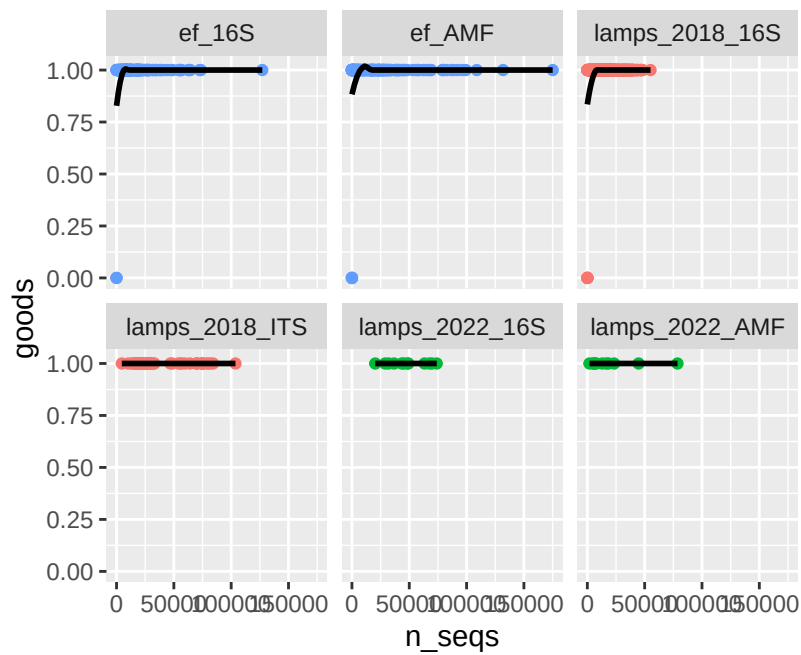
Read Count Comparison Across Projects



\$summary_line



\$goods_cov



```

> p_iNEXt_list <- function(physeq_obj, q = c(0, 1, 2),
+   nCores = 1, type = 1) {
+   plot_title <- as_label(enexpr(physeq_obj)) %>% str_replace .... [TRUNCATED]

> load("data/output/rdata/rarecurves.rda")

> cat("\n", rep("=", 40), "\n")

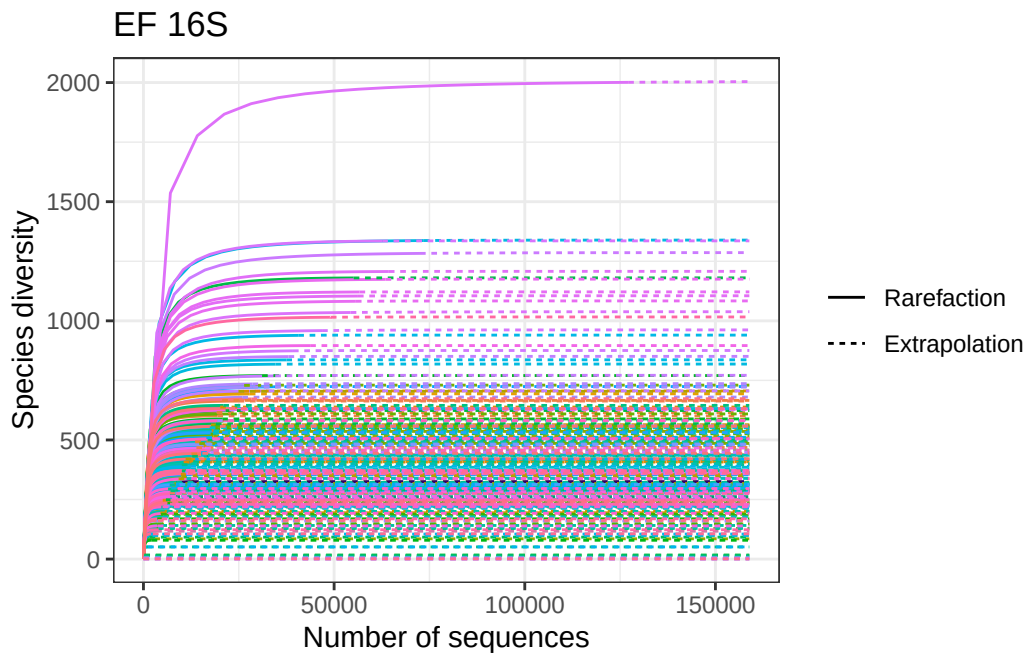
= = = = =

> cat("Rarefaction curves\n")
Rarefaction curves

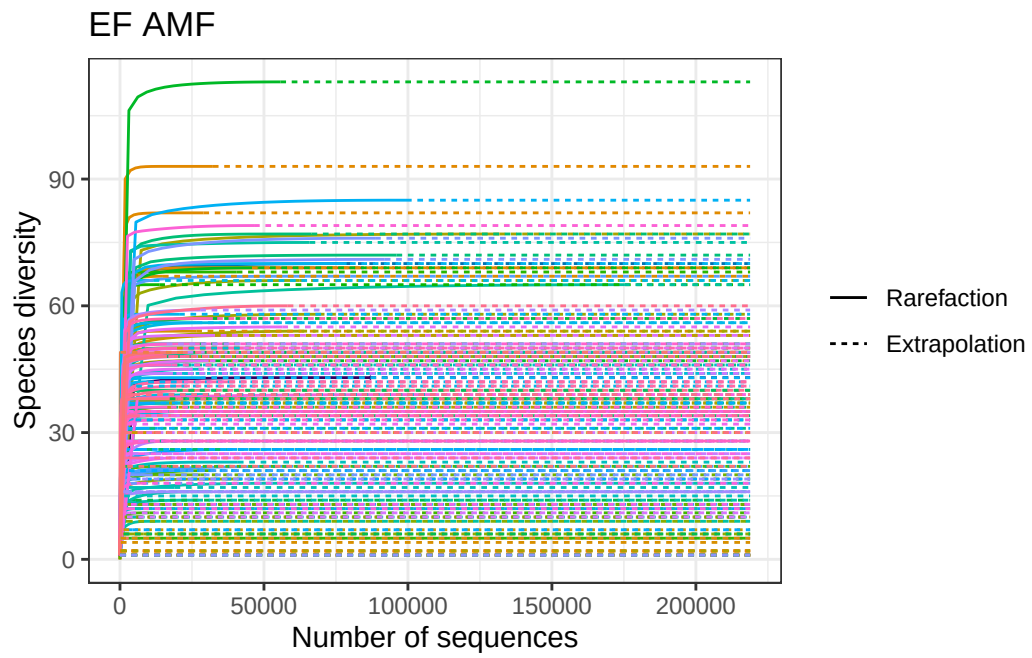
> cat(rep("=", 40), "\n")
= = = = =

> rarecurve_results %>% purrr::imap(function(project_list,
+   project_name) {
+   purrr::imap(project_list, function(inext_results, inext_name) .... [TRUNCATED]
$mxg_ef
$mxg_ef$ef_16S_iNEXT

```



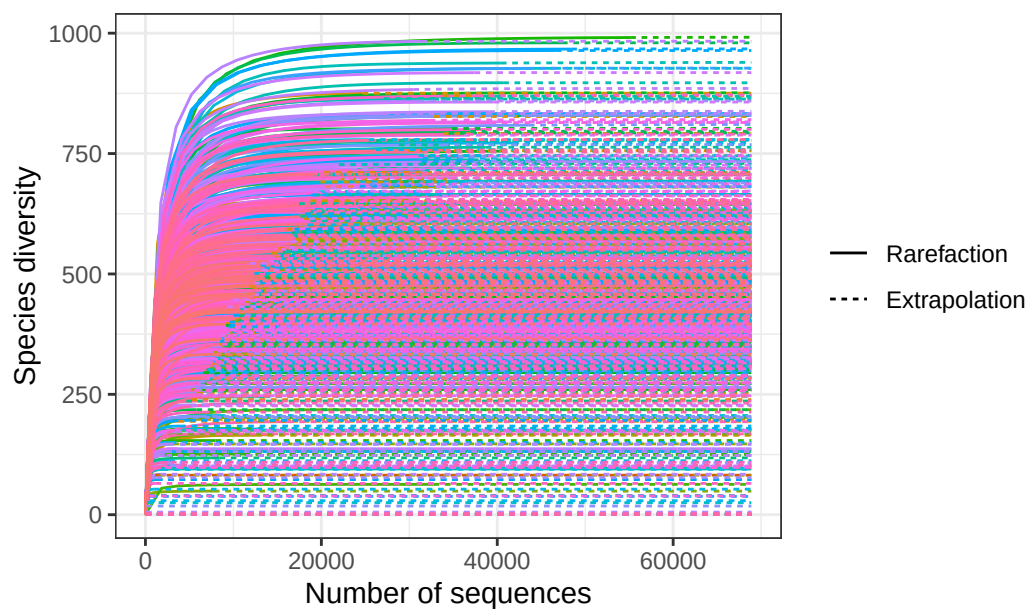
```
$mxg_ef$ef_AMF_iNEXT
```



```
$mxg_lamps_2018
```

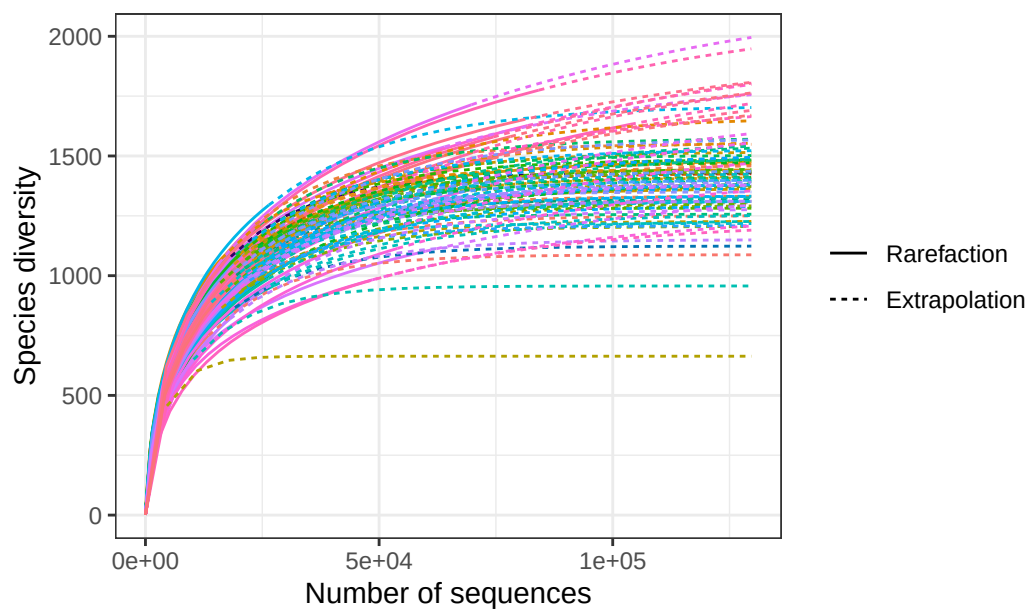
```
$mxg_lamps_2018$lamps_16S_2018
```

LAMPS 16S 2018

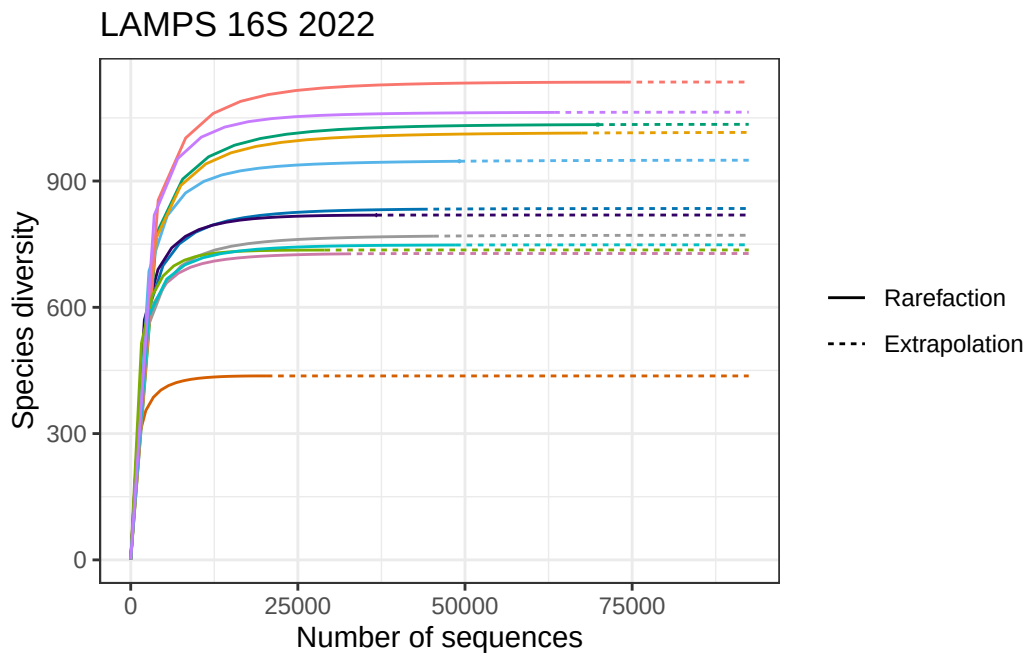


`$mxg_lamps_2018$lamps_ITS_2018`

LAMPS ITS 2018

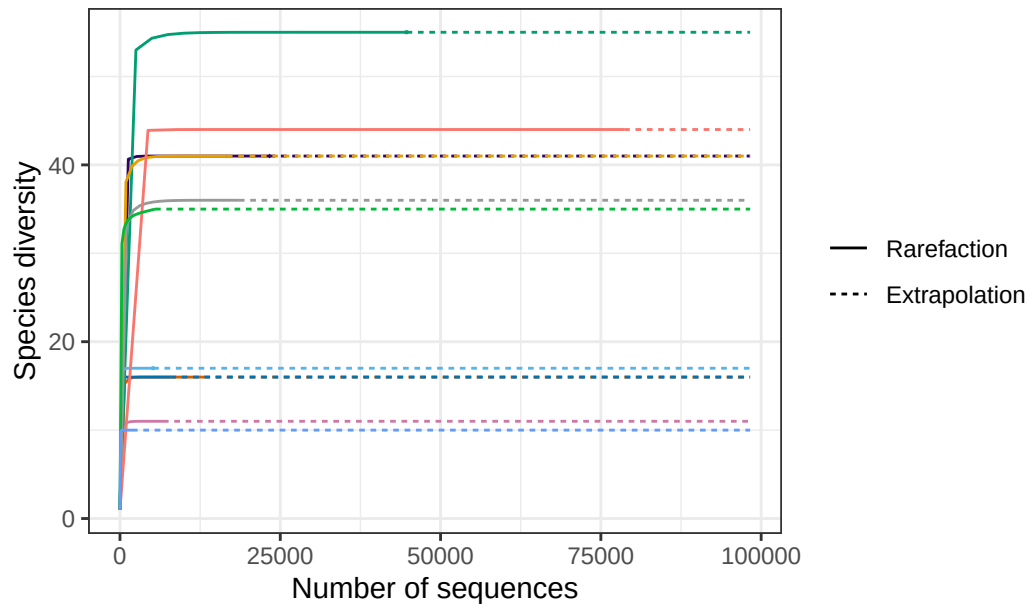


\$mxg_lamps_2022
\$mxg_lamps_2022\$lamps_16S_2022



\$mxg_lamps_2022\$lamps_AMF_2022

LAMPS AMF 2022

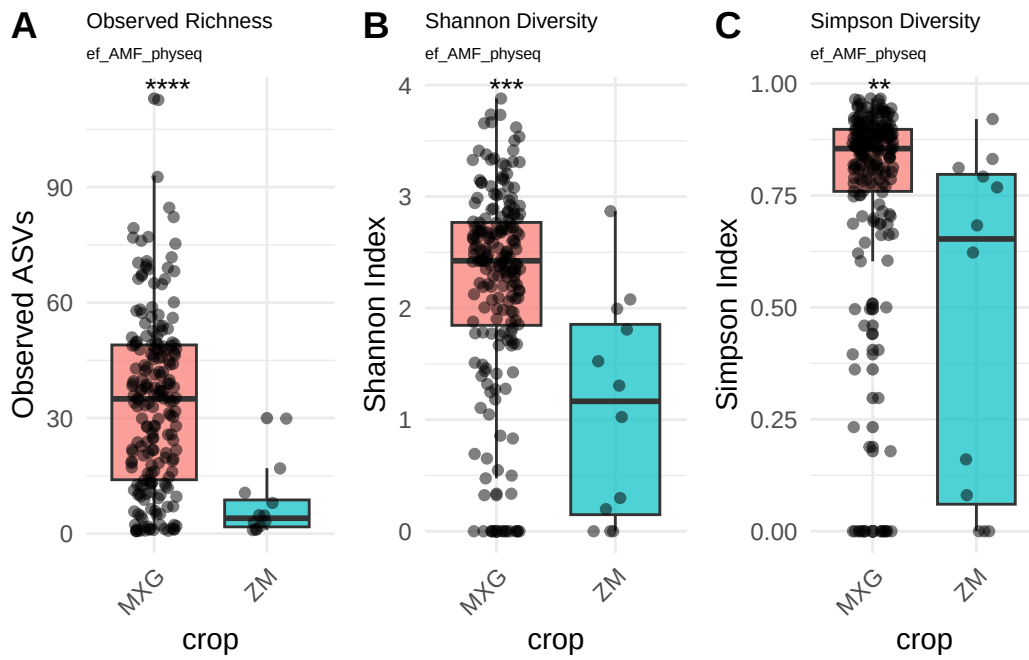
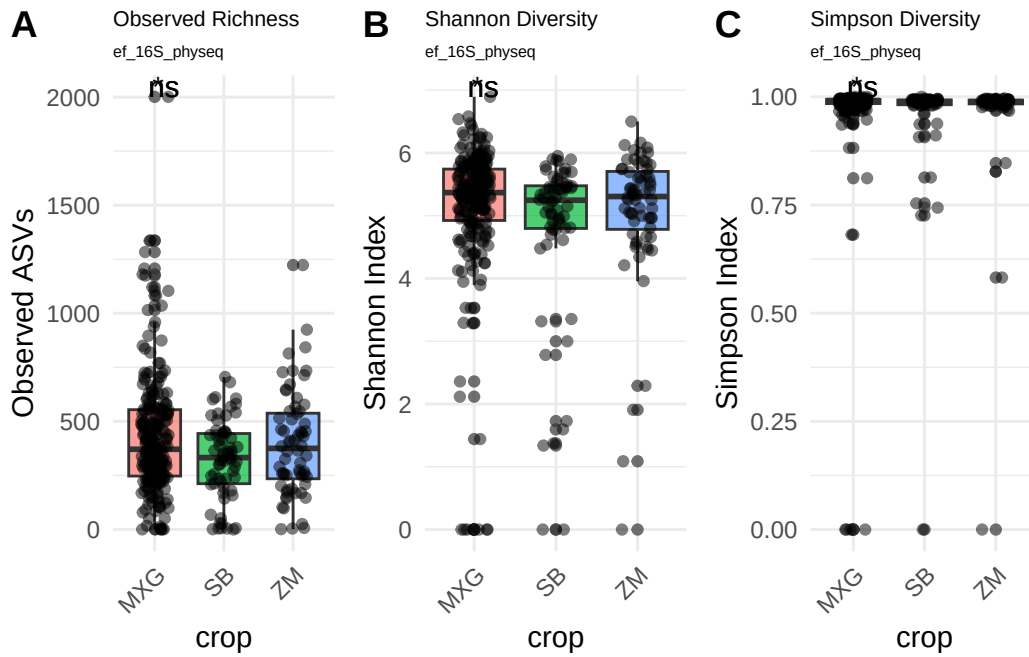


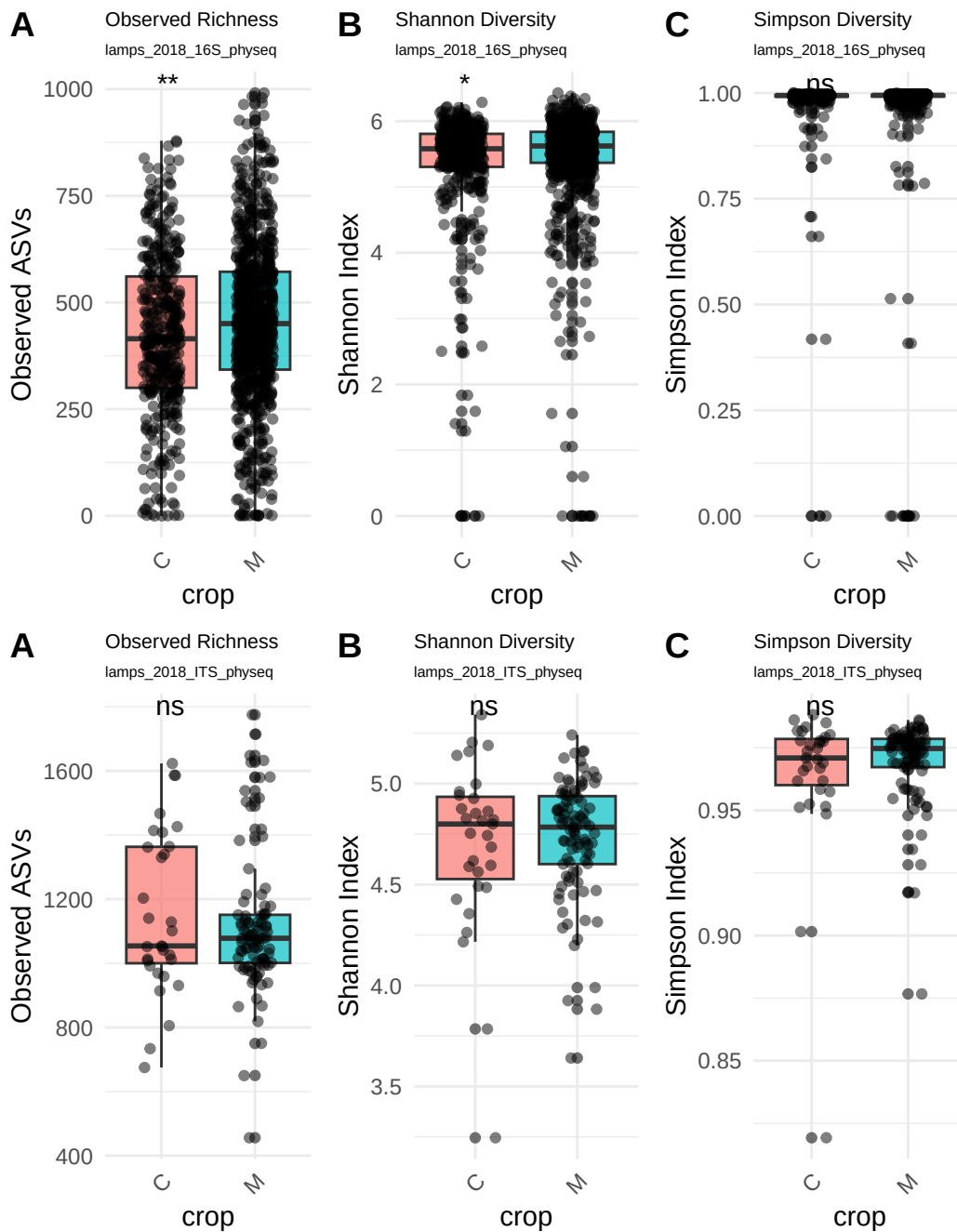
Section 2 — Alpha Diversity Analyses

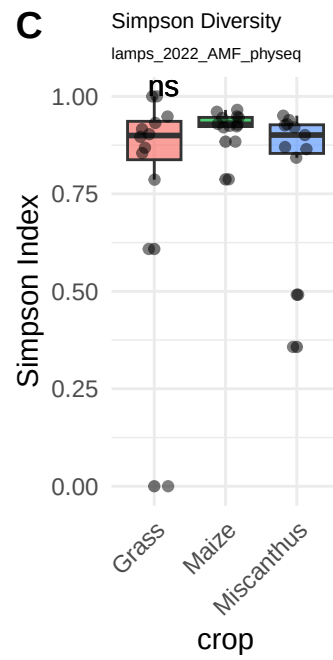
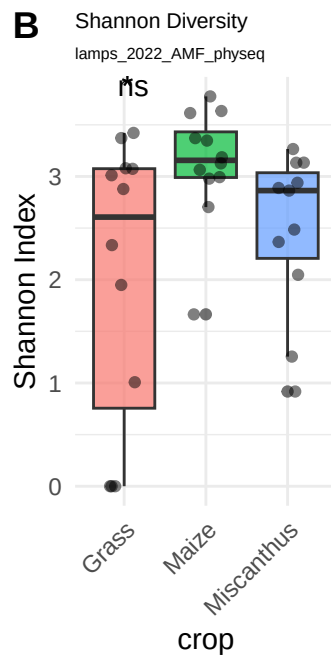
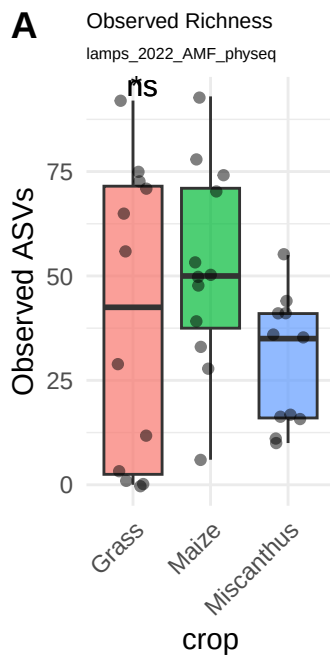
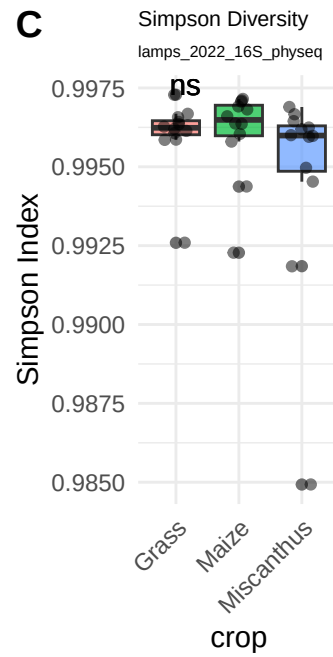
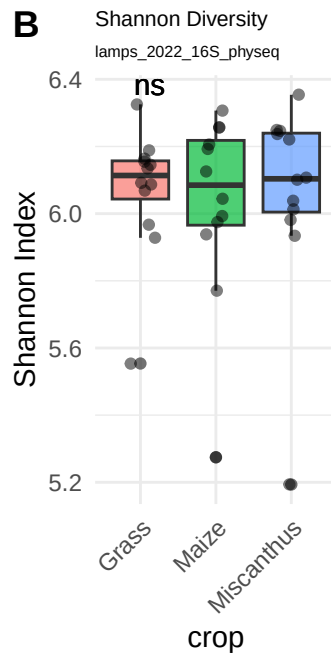
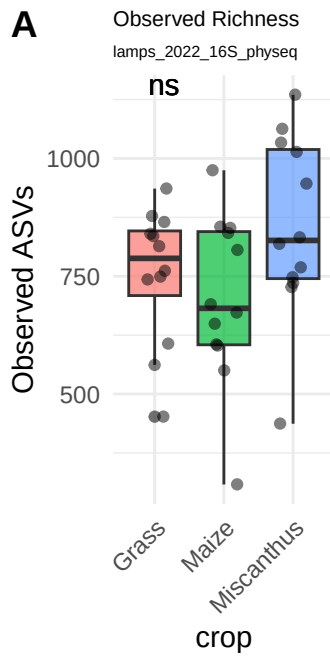
This section runs `05_alpha_beta_div.R`, including:

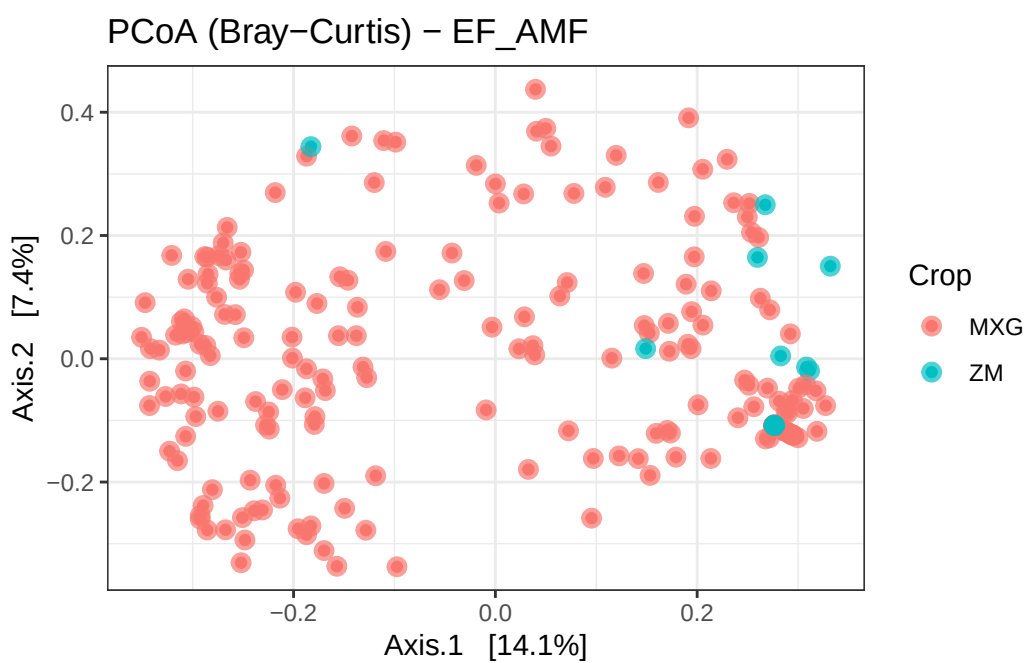
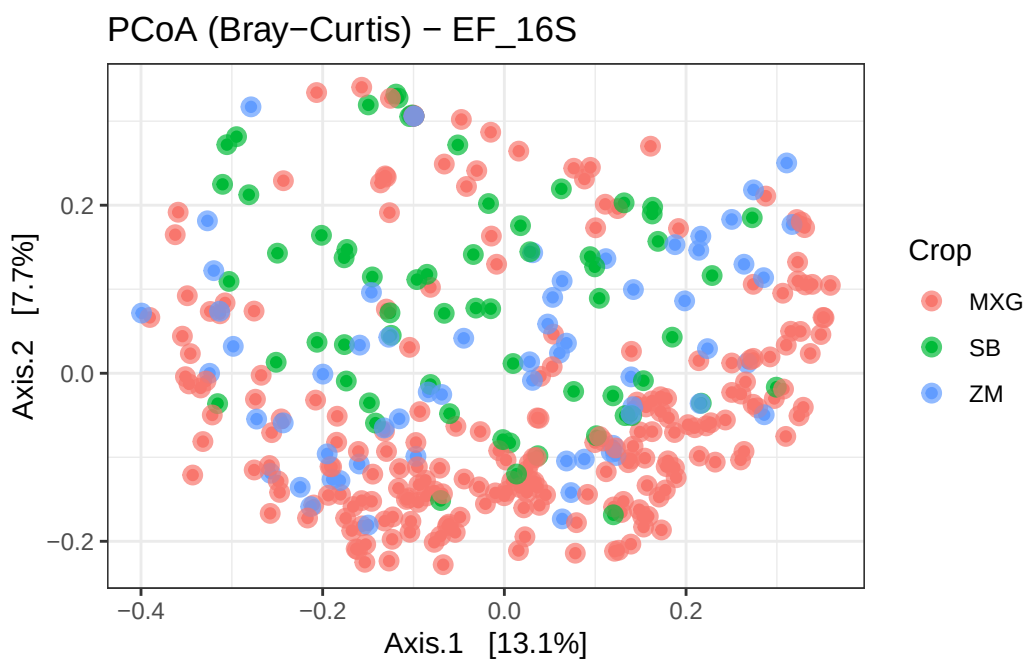
- Richness and alpha diversity summaries
- Beta diversity ordinations

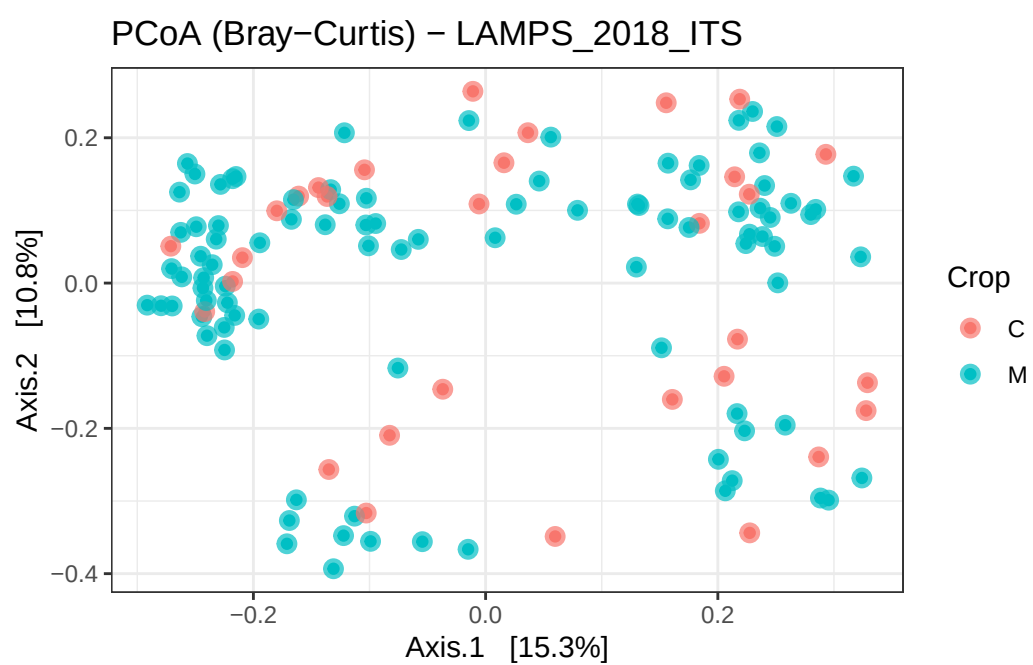
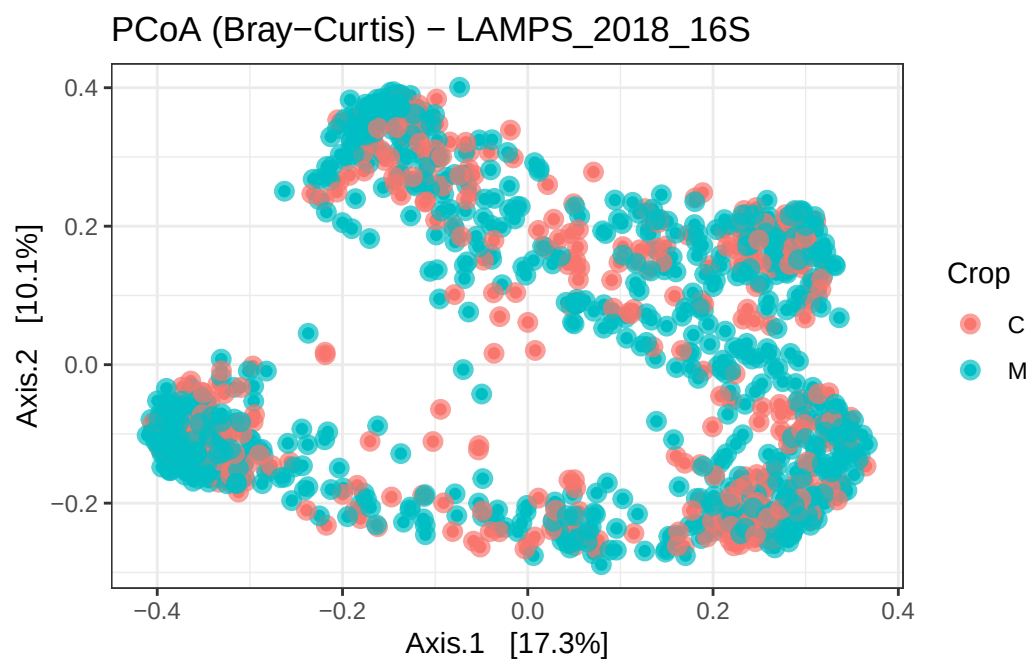
```
source("R/05_alpha_beta_div.R", local = knitr::knit_global())
```

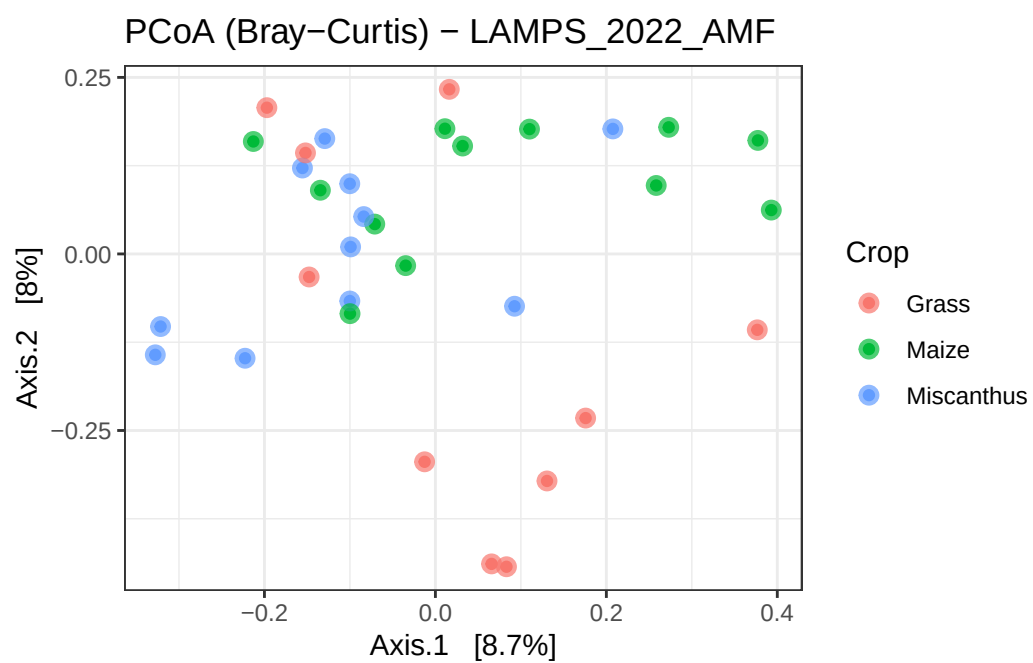
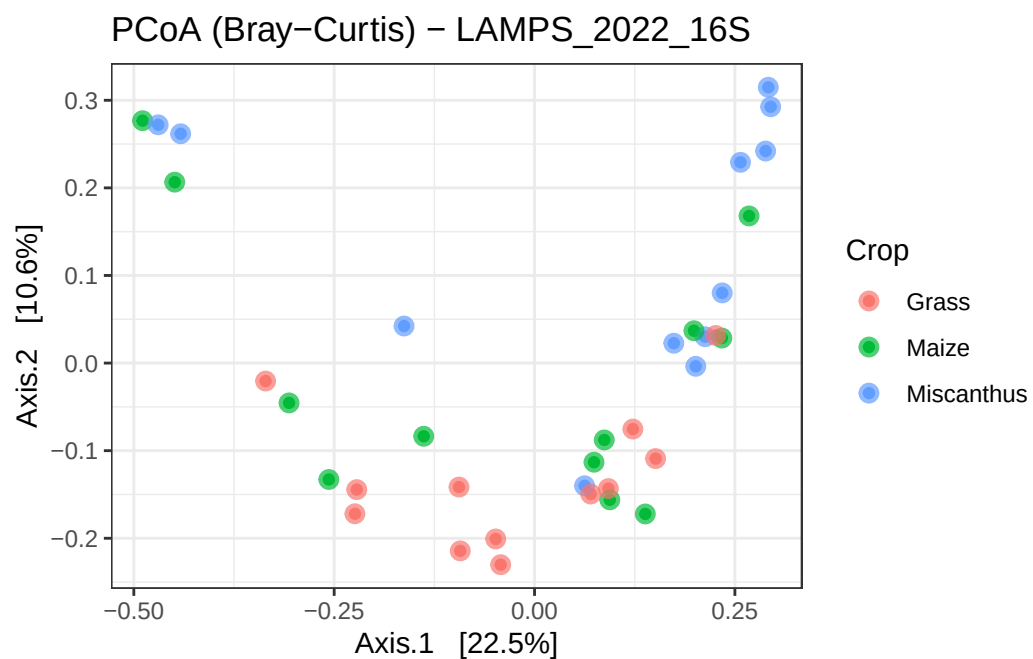













```

=====
Alpha Diversity by Project and Crop
=====
$ef_physeq_list

```

\$ef_physeq_list\$ef_16S_physeq
\$ef_physeq_list\$ef_16S_physeq\$all_div_plots

\$ef_physeq_list\$ef_AMF_physeq
\$ef_physeq_list\$ef_AMF_physeq\$all_div_plots

\$lamps_2018_physeq_list
\$lamps_2018_physeq_list\$lamps_2018_16S_physeq
\$lamps_2018_physeq_list\$lamps_2018_16S_physeq\$all_div_plots

\$lamps_2018_physeq_list\$lamps_2018_ITS_physeq
\$lamps_2018_physeq_list\$lamps_2018_ITS_physeq\$all_div_plots

\$lamps_2022_physeq_list
\$lamps_2022_physeq_list\$lamps_2022_16S_physeq
\$lamps_2022_physeq_list\$lamps_2022_16S_physeq\$all_div_plots

\$lamps_2022_physeq_list\$lamps_2022_AMF_physeq
\$lamps_2022_physeq_list\$lamps_2022_AMF_physeq\$all_div_plots

= = = = =
Beta Diversity Project and Crop
= = = = =

\$ef_physeq_list
\$ef_physeq_list\$ef_16S_physeq

\$ef_physeq_list\$ef_AMF_physeq

\$lamps_2018_physeq_list
\$lamps_2018_physeq_list\$lamps_2018_16S_physeq

\$lamps_2018_physeq_list\$lamps_2018_ITS_physeq

```
$lamps_2022_physeq_list
$lamps_2022_physeq_list$lamps_2022_16S_physeq

$lamps_2022_physeq_list$lamps_2022_AMF_physeq
```

```

= = = = =
Alpha Diversity Summary by Project and Crop
= = = = =
# A tibble: 15 x 10
  project region crop      n mean_observed sd_observed mean_shannon sd_shannon
  <chr>    <chr> <chr> <int>      <dbl>      <dbl>      <dbl>      <dbl>
1 ef_phys~ 16S   MXG    223      429.        284.        5.13      1.17
2 ef_phys~ 16S   SB      64      323.        183.        4.75      1.39
3 ef_phys~ 16S   ZM      64      394.        242.        5.05      1.13
4 ef_phys~ AMF   MXG    200       33.7        22.4        2.22      0.911
5 ef_phys~ AMF   ZM      12        7.25         8.62        1.09      0.986
6 lamps_2~ 16S   C     388      424.        190.        5.34      0.963
7 lamps_2~ 16S   M     788      455.        191.        5.43      0.864
8 lamps_2~ ITS   C      31     1152.        252.        4.69      0.425
9 lamps_2~ ITS   M      93     1118.        223.        4.73      0.296
10 lamps_2~ 16S   Grass    12      754.        144.        6.07      0.191
11 lamps_2~ 16S   Maize    12      701.        179.        6.03      0.286
12 lamps_2~ 16S   Misc~    12      855.        194.        6.06      0.301
13 lamps_2~ AMF   Grass    12       39.8        35.5        2.01      1.39
14 lamps_2~ AMF   Maize    12       51.8        24.1        3.12      0.555
15 lamps_2~ AMF   Misc~    11       29.3        15.6        2.48      0.783
# i 2 more variables: mean_simpson <dbl>, sd_simpson <dbl>

```

Session information

```
sessionInfo()
```

```
R version 4.4.2 (2024-10-31)
Platform: x86_64-pc-linux-gnu
Running under: Arch Linux
```


Matrix products: default

BLAS/LAPACK: /usr/lib/libopenblas.so.0.3; LAPACK version 3.12.0

locale:

[1] LC_CTYPE=C.UTF-8	LC_NUMERIC=C	LC_TIME=C.UTF-8
[4] LC_COLLATE=C.UTF-8	LC_MONETARY=C.UTF-8	LC_MESSAGES=C.UTF-8
[7] LC_PAPER=C.UTF-8	LC_NAME=C	LC_ADDRESS=C
[10] LC_TELEPHONE=C	LC_MEASUREMENT=C.UTF-8	LC_IDENTIFICATION=C

time zone: America/Chicago

tzcode source: system (glibc)

attached base packages:

[1] stats graphics grDevices datasets utils methods base

other attached packages:

[1] iNEXT_3.0.2	readxl_1.4.5	ggtext_0.1.2
[4] metagMisc_0.6.0.9000	microbiome_1.28.0	janitor_2.2.1
[7] data.table_1.17.8	lubridate_1.9.4	forcats_1.0.1
[10] stringr_1.5.1	dplyr_1.1.4	purrr_1.1.0
[13] readr_2.1.5	tidyr_1.3.1	tibble_3.2.1
[16] ggplot2_3.5.2	tidyverse_2.0.0	vegan_2.7-1
[19] permute_0.9-8	phyloseq_1.50.0	conflicted_1.2.0

loaded via a namespace (and not attached):

[1] RColorBrewer_1.1-3	jsonlite_2.0.0	magrittr_2.0.3
[4] farver_2.1.2	rmarkdown_2.29	fs_1.6.6
[7] zlibbioc_1.52.0	vctrs_0.6.5	multtest_2.62.0
[10] memoise_2.0.1	rstatix_0.7.3	tinytex_0.54
[13] htmltools_0.5.8.1	DEoptim_2.2-8	broom_1.0.10
[16] cellranger_1.1.0	Rhdf5lib_1.28.0	Formula_1.2-5
[19] rhdf5_2.50.2	parallelly_1.45.1	plyr_1.8.9
[22] cachem_1.1.0	igraph_2.1.4	lifecycle_1.0.4
[25] iterators_1.0.14	pkgconfig_2.0.3	Matrix_1.7-2
[28] R6_2.6.1	fastmap_1.2.0	GenomeInfoDbData_1.2.13
[31] future_1.67.0	snakecase_0.11.1	digest_0.6.37
[34] numDeriv_2016.8-1.1	colorspace_2.1-1	S4Vectors_0.44.0
[37] rprojroot_2.1.1	ggpubr_0.6.2	labeling_0.4.3
[40] clusterGeneration_1.3.8	timechange_0.3.0	httr_1.4.7
[43] abind_1.4-8	mgcv_1.9-1	compiler_4.4.2
[46] here_1.0.2	withr_3.0.2	doParallel_1.0.17
[49] S7_0.2.0	backports_1.5.0	optimParallel_1.0-2
[52] carData_3.0-5	maps_3.4.3	ggsignif_0.6.4

[55]	MASS_7.3-64	scatterplot3d_0.3-44	biomformat_1.34.0
[58]	tools_4.4.2	ape_5.8-1	phytools_2.5-2
[61]	future.apply_1.20.0	glue_1.8.0	quadprog_1.5-8
[64]	nlme_3.1-167	rhdf5filters_1.18.1	gridtext_0.1.5
[67]	grid_4.4.2	Rtsne_0.17	cluster_2.1.6
[70]	reshape2_1.4.4	ade4_1.7-23	generics_0.1.4
[73]	gtable_0.3.6	tzdb_0.5.0	hms_1.1.3
[76]	utf8_1.2.4	xml2_1.4.0	car_3.1-3
[79]	XVector_0.46.0	BiocGenerics_0.52.0	foreach_1.5.2
[82]	pillar_1.10.1	splines_4.4.2	lattice_0.22-6
[85]	renv_1.1.5	survival_3.7-0	tidyselect_1.2.1
[88]	Biostrings_2.74.1	knitr_1.49	IRanges_2.40.1
[91]	stats4_4.4.2	xfun_0.53	expm_1.0-0
[94]	Biobase_2.66.0	stringi_1.8.7	UCSC.utils_1.2.0
[97]	geodist_0.1.1	yaml_2.3.10	evaluate_1.0.3
[100]	codetools_0.2-20	BiocManager_1.30.26	cli_3.6.5
[103]	Rcpp_1.0.14	GenomeInfoDb_1.42.3	globals_0.18.0
[106]	coda_0.19-4.1	parallel_4.4.2	listenv_0.9.1
[109]	phangorn_2.12.1	scales_1.4.0	crayon_1.5.3
[112]	combinat_0.0-8	rlang_1.1.6	cowplot_1.2.0
[115]	fastmatch_1.1-6	mnormt_2.1.1	